

STILLWATER 5 WNW, OK, USA

WMO: 746430

Lat: 36.1346N

Lon: 97.1082W

Elev: 888

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	15.8	1.6	6.2	13.9	6.5	7.9	18.9	16.8	37.4	15.0	36.0	4.2	N/A	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	100.5	74.0	97.0	74.9	94.0	75.2	79.1	91.7	78.0	90.6	77.1	89.3	4.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.7	138.9	85.7	74.6	133.6	84.5	73.5	128.6	83.3	43.2	91.7	42.1	90.5	41.0	89.6	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.0	12.0	10.2	1.5	103.5	7.7	4.6	-4.0	106.8	-8.6	109.5	-12.9	112.1	-18.5	115.5	
			-0.6	80.5	9.5	1.6	-7.4	81.6	-12.9	82.6	-18.2	83.4	-25.1	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	36.7	39.7	50.9	59.2	68.0	77.4	81.3	80.3	72.7	60.1	48.7	38.6
	DBStd	18.06	10.27	11.29	10.73	9.24	8.10	5.28	5.31	5.71	6.95	9.40	9.51	9.58
	HDD50	1414	426	320	118	24	2	0	0	0	0	20	134	370
	HDD65	3862	877	708	450	220	65	1	0	0	16	210	496	819
	CDD50	4908	15	33	147	300	560	822	970	939	680	332	94	16
	CDD65	1880	0	1	14	47	157	373	505	474	246	57	6	0
	CDH74	21700	4	19	151	516	1560	4186	6315	5642	2601	642	62	2
	CDH80	10348	0	3	26	128	514	1895	3449	3026	1119	180	8	0
Wind	WSAvg	4.0	4.5	4.9	5.2	5.3	4.1	3.6	2.9	3.0	3.1	3.6	4.1	4.2
Precipitation	PrecAvg	34.6	1.2	1.4	2.7	3.1	4.9	4.3	3.0	3.1	4.1	3.0	2.3	1.6
	PrecMax	46.4	3.7	5.4	7.7	8.3	14.6	10.6	8.2	8.5	12.4	8.7	6.7	5.1
	PrecMin	19.7	0.0	0.1	0.0	0.2	0.6	0.8	0.0	0.0	0.4	0.0	0.1	0.1
	PrecStd	6.7	1.1	1.2	1.9	1.9	2.8	2.4	1.9	2.1	2.6	2.1	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	76.8	83.7	88.6	92.2	99.3	105.5	106.8	98.3	88.9	80.5	70.8
		MCWB	54.2	58.9	63.9	66.2	70.5	74.7	72.8	73.0	71.4	71.2	63.9	56.4
	2%	DB	66.6	71.0	78.1	83.7	88.8	95.0	101.2	100.6	92.7	85.1	75.1	65.2
		MCWB	52.0	56.7	60.9	64.1	71.1	75.2	73.6	74.7	73.4	68.1	59.9	54.6
	5%	DB	61.4	65.4	73.9	79.9	85.9	91.9	97.4	96.9	90.0	81.3	70.7	60.1
		MCWB	50.2	54.7	58.8	63.2	70.4	75.3	74.9	75.2	73.2	66.0	58.2	50.4
	10%	DB	55.5	60.0	69.6	76.5	82.4	89.4	94.1	93.5	86.7	77.2	66.2	56.1
		MCWB	45.7	50.1	56.7	63.0	69.5	74.6	75.9	74.6	72.0	64.8	55.5	48.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	62.8	67.6	70.6	76.5	80.0	80.8	80.2	77.5	74.1	67.8	62.9
		MCDB	64.9	70.6	80.3	80.4	86.5	92.6	92.5	94.0	89.7	84.5	73.3	68.2
	2%	WB	53.2	60.0	64.7	68.2	74.5	78.4	79.2	78.6	76.2	72.1	62.7	56.7
		MCDB	62.4	68.6	74.9	78.4	84.7	90.4	91.1	92.5	88.1	81.1	69.8	59.6
	5%	WB	49.7	56.6	62.1	66.7	72.8	77.3	78.1	77.4	75.1	70.1	60.2	52.5
		MCDB	59.7	63.9	71.3	76.7	82.5	89.0	90.4	90.7	86.5	77.8	67.5	58.6
	10%	WB	46.1	51.0	58.6	64.8	71.1	76.1	77.1	76.2	73.8	67.4	57.2	48.2
		MCDB	54.1	60.6	66.9	73.9	80.0	87.7	89.5	88.5	84.2	74.0	64.2	54.6
Mean Daily Temperature Range	5% DB	MDBR	24.5	23.6	24.9	24.0	21.3	20.4	21.7	22.4	23.4	24.6	25.1	23.0
		MCDBR	35.7	33.3	30.3	28.9	24.3	23.4	26.6	27.6	26.3	28.7	31.0	31.5
		MCWBR	21.8	20.2	17.1	15.5	10.9	8.4	7.3	7.9	9.6	13.6	17.4	21.1
	5% WB	MCDBR	32.4	29.2	25.8	23.3	20.3	19.6	20.4	21.5	20.8	20.5	23.6	25.0
		MCWBR	21.3	19.9	15.2	14.1	10.6	8.5	8.1	8.3	8.3	11.9	16.3	19.5
Clear-Sky Solar Irradiance	taub	0.297	0.302	0.342	0.382	0.419	0.429	0.434	0.436	0.393	0.348	0.313	0.291	
	taud	2.496	2.493	2.411	2.315	2.283	2.291	2.316	2.253	2.357	2.469	2.504	2.523	
	Ebn at Noon	287	299	294	286	275	271	269	266	273	278	279	283	
	Edh at Noon	26	29	35	40	42	42	41	43	36	30	25	24	
All-Sky Solar Radiation	RadAvg	850	1050	1370	1705	1872	2138	2140	1936	1610	1211	902	718	
	RadStd	84	157	88	118	189	189	118	118	133	149	113	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page